

Extreme Prejudice: Sentencing Spillovers from Murder Cases

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Judgment is contextual: exposure to extreme cases can bias subsequent evaluations. I document such a bias in 2.6 million criminal sentencing decisions from Cook County and Pennsylvania, using two complementary designs. First, on the day of a murder sentence, judges issue 11–16% longer sentences to unrelated defendants. Effects persist for two weeks for some cases. Second, judges who rotate onto Philadelphia’s homicide docket issue 20% longer sentences for the entire rotation and three months afterward. Effects are larger for severe cases and minority defendants, suggesting a salience mechanism.

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Decision-makers frequently evaluate cases in sequence: recruiters screen resumes, buyers tour homes, and judges sentence defendants. Sequential evaluation can bias judgment because it invites comparison: memorable cases may become benchmarks that change how subsequent cases are assessed. For example, consider a judge who has recently sentenced a particularly extreme case such as a murder. If subsequent crimes remind her of the murder, she may become harsher (an assimilation effect). However, if subsequent crimes appear run-of-the-mill in contrast to the murder, she may become more lenient (a contrast effect).

In this paper, I study how exposure to extreme cases affects decisions. I analyze 2.6 million sentencing decisions made by state trial court judges in Cook County, Illinois (2011–2024) and Pennsylvania (1991–2024) using two complementary designs. In the first design, I estimate the effect of sentencing a murder on subsequent sentencing. In the second design, I estimate how sentencing evolves before, during, and after assignment to Philadelphia’s homicide docket. Together, these designs show how episodic and prolonged exposure to extreme cases can bias judgments in a high-stakes setting with experienced decision-makers.

The murder design compares treated judges who sentence a murder to control judges in the same courthouse (Cook County) or judicial district (Pennsylvania, excluding Philadelphia) who do not, using stacked difference-in-differences (Deshpande and Li, 2019; Wing et al., 2024). I consider the jurisdictions separately because each has unique advantages. In Cook County, cases are randomly assigned to judges within a courthouse, helping to allay concerns about unobservable sorting. In Pennsylvania, I observe a much larger sample that contains fine-grained information on defendant history and crime severity.

The murder design requires two identifying assumptions. First, judges who do not sentence a murder must provide a valid counterfactual for those who do. I provide evidence for this assumption by showing that sentencing for treated and control judges is both similar in level and evolves along parallel trends conditional on observables. Second, the sentencing of a murder must not cause sorting of cases on unobservables. Some concern is warranted here: judges sentence high-severity cases 2–5 percentage points more often on the day of a murder sentence, even though I exclude murder co-defendants from the analysis. However, conditional on offense severity, I do not detect additional sorting on defendant demographics or other indicators of case severity including initial charge class, case length, and the application of mandatory minimums. Oster (2019) bounds imply that unobservable sorting would have to be at least 14 times larger than observable sorting to explain the results.

I find evidence of assimilation effects: following a murder sentence, judges become harsher. In Cook County, sentences are 16% longer on the day of a murder sentence (139 days compared to a mean of 885), and in Pennsylvania sentences are 11% longer (14 days compared to a mean of 123; sentences are shorter in Pennsylvania because the sample in-

cludes misdemeanors). Decomposing this result into intensive and extensive margins, the bias causes judges to issue longer sentences to defendants who would be incarcerated regardless, but does not affect the probability of incarceration. Sentences remain elevated for two weeks in Pennsylvania and for two weeks for severe cases in Cook County. Because judges regularly sentence murders, a meaningful fraction of high-severity cases are subject to bias on an ongoing basis (10% in Cook County; 2% in Pennsylvania).

The homicide docket design estimates the effect of prolonged exposure to extreme cases. I leverage the fact that case assignment in Philadelphia, the largest judicial district in Pennsylvania, uses a docket system: when judges are assigned to the homicide docket, their exposure to murder cases sharply rises and can remain elevated for years. Importantly for identification, even after judges rotate onto the homicide docket, they continue to handle cases across the severity spectrum. I compare the handling of non-murder cases between treated judges who rotate onto the homicide docket and control judges who are never assigned to the homicide docket using a stacked difference-in-differences design. Because the length of a homicide docket rotation is not standardized, I estimate the effects of rotating onto and off of the homicide docket separately.

The homicide docket design requires two identifying assumptions. First, judges who do not rotate onto the homicide docket must provide a valid counterfactual for those who do. To assuage this concern, I show that sentencing for treated and control judges evolves along parallel trends conditional on observables. Second, rotating onto the homicide docket must not coincide with being assigned non-murder cases that are more severe in an unobservable way. While I cannot directly rule out unobservable sorting, I control for Pennsylvania’s fine-grained sentencing guideline cells which reflect the legally relevant factors that determine sentencing ranges including criminal history and charge severity. Oster (2019) bounds imply that unobservable sorting would have to be at least 10 times larger than observable sorting to explain the results.

Rotating onto the homicide docket causes non-murder sentence lengths to rise by 20% (125 days compared to a mean of 617). Effects persist for the entire length of the docket rotation (median: 2.7 years) and three months afterward. In the three years after judges rotate off the homicide docket, sentences fall by 80 days, reversing 64% of the rotate-on treatment effect. I decompose the overall effect into intensive and extensive margins. Conditional on incarceration, rotating onto the homicide docket raises sentences by 14% (143 days compared to a mean of 1,005), and rotating off reverses 76% of the effect. I do not detect effects on incarceration probability.

To explore mechanisms, I estimate heterogeneous treatment effects using triple-differences.

Across both jurisdictions and designs, treatment effects are larger for high-severity cases.¹ In Cook County and Philadelphia, effects are larger for minorities. Together with the fact that effects fade over time, these results are consistent with a memory mechanism in which salient cases distort subsequent sentencing, because race and severity are dimensions along which defendants resemble the typical murder defendant (Bordalo et al., 2015).

The differential treatment effects by race present meaningful equity concerns. While a salience-driven mechanism could be race-neutral in principle, in practice over 90% of the convicted murderers I study in Cook County and Pennsylvania are minorities. In Cook County, the minority-White sentencing gap doubles on the day of a murder sentencing.² In Philadelphia, the effect of the homicide docket on minorities is 2.4 times larger than the effect on White defendants (144 vs. 59 days).

This paper contributes to a large literature focused on how extraneous factors can bias judicial decisions.³ I study how judges respond to violent events that they are exposed to within the normal course of their job, but this treatment variation is conceptually similar to a strand of the literature focused on violent external events including terrorist attacks (Shayo and Zussman, 2011; Brodeur and Wright, 2019; McConnell and Rasul, 2021; Emeriau, 2024) and high profile killings (Hanemaaijer et al., 2026). Consistent with these papers, I find that effects are largest for salient subpopulations.

Within the extraneous factors literature, this paper is most closely related to the strand focused on how a decision in one case can spill over to affect another.⁴ Early lab experiments

¹In Cook County, where I only observe felonies, I define the higher X and 1 class felonies as high-severity. In Pennsylvania, where I observe felonies and misdemeanors, I define felonies as high-severity.

²Conditional on initial and final charge class, the minority-White sentencing gap in Cook County is 177 days compared to a treatment effect of 185 days.

³Judges can be influenced by factors external to the courtroom including labor market conditions (Ichino et al., 2003), media coverage (Lim et al., 2015; Philippe and Ouss, 2018; Ash and Poyker, 2024), elections (Huber and Gordon, 2004; Dyke, 2007; Lim, 2013; Berdejó and Yuchtman, 2013; Berdejó and Chen, 2017; Chen, 2024), constituent composition (Feigenberg and Miller, 2021), norms (Abrams et al., 2022; Chen and Philippe, 2023), and immigration flows (Bertoli et al., 2023). Judges may also be sensitive to factors which influence their internal state including hunger (Danziger et al. (2011), but see Weinshall-Margel and Shapard (2011); Glöckner (2016)), sleep deprivation (Cho et al. (2016), but see Spamann (2018)), mood (Eren and Mocan, 2018; Chen and Loecher, 2025), temperature (Heyes and Saberian (2019); Craigie et al. (2023); Behrer and Bolotnyy (2024), but see Spamann (2022); Evans and Siminski (2024)), and air pollution (Sarmiento and Nowakowski (2023), but see Hou and Wang (2020)). A closely related literature documents the effects of defendant and judge characteristics. A defendant’s gender (Butcher et al., 2017), race (Abrams et al., 2012; Arnold et al., 2022), and ethnicity (Gazal-Ayal and Sulitzeanu-Kenan, 2010; Shayo and Zussman, 2011, 2017) can affect decisions. The literature on judge characteristics is more mixed (Lim et al., 2016; Harris and Sen, 2019; Cohen and Yang, 2019). Multiple studies document effects of the interaction between judge and defendant characteristics in the form of in-group bias (Depew et al., 2017; Ahrsjö et al., 2024; Ash et al., 2025).

⁴More broadly, a literature in psychology and economics considers biases that arise in the context of sequential decisions (Beebe-Center (1929); Sherif et al. (1958); Tversky and Kahneman (1974), and see Furnham and Boo (2011) for a literature review). This paper joins a recent strand documenting bias in the

in the context of judicial decisions found evidence of contrast effects (Pepitone and DiNubile, 1976). More recent field studies of judges also find evidence consistent with contrast effects: Chen et al. (2016) document evidence of the gambler’s fallacy among asylum judges, finding that a prior approval increases the probability of a subsequent denial by three percentage points. Leibovitch (2016) finds contrast effects with respect to historical caseload severity among judges in Pennsylvania. However, Bindler and Hjalmarsson (2019) find assimilation effects among jury decisions in London’s Old Bailey Criminal Court: a prior guilty verdict increases the chance of a subsequent guilty verdict by five percentage points.

This paper contributes to the literature on spillover effects by showing that exposure to extreme cases can persistently bias sentencing for weeks to months after exposure ends. Moreover, the direction of the effects is surprising, given that the most closely related lab (Pepitone and DiNubile, 1976) and field (Chen et al., 2016) studies of judges find effects that go in the opposite direction.⁵ Indeed, these results are the first evidence of assimilation spillover effects among judges.⁶

This paper also relates to a smaller strand of the literature focused on how institutional design shapes judicial decisions. Judicial decisions are affected by constraints on judicial discretion (Fischman and Schanzenbach, 2012), caseloads (Engel and Weinshall, 2020; Shumway and Wilson, 2022), retirement policies (Ash and MacLeod, 2024), and the threat of reversals (Bhuller and Sigstad, 2025). This paper contributes by highlighting the role of case assignment: I find that effects are larger and more persistent when extreme cases are concentrated due to docket-based assignment relative to more evenly distributed assignment mechanisms. Unlike many sources of variation studied in the extraneous factors literature (e.g., terrorist attacks, media coverage), case assignment is a policy-relevant margin because it sits directly under the purview of the court, and making changes to case assignment does not necessarily introduce additional costs.

The remainder of the paper proceeds as follows. Section 1 describes the data and provides institutional details of the Cook County and Pennsylvania judicial systems. Section 2 de-

field. Evidence of anchoring in the field has been found in art auctions, house sales, sports betting, game shows, and procurement auctions (Beggs and Graddy, 2009; Bucchianeri and Minson, 2013; McAlvanah and Moul, 2013; Jetter and Walker, 2017; Ferraro et al., 2022). Evidence of contrast effects in the field has been found in commuting patterns, speed dating, and financial markets (Simonsohn, 2006; Bhargava and Fisman, 2014; Hartzmark and Shue, 2018).

⁵In comparison to Chen et al. (2016), I study a different setting (criminal rather than asylum courts) and a different margin (continuous sentence length rather than binary approval decisions).

⁶Some early lab studies find evidence of judges anchoring to prosecutorial demands (Englich and Mussweiler, 2001) or a number introduced in a prior motion (Guthrie et al., 2000), but do not study spillovers across cases. Though not a standard assimilation effect, Angelova et al. (2025) document a pattern that is similar in spirit: bail judges temporarily lower release rates after presiding over an unrelated case in which a defendant released by another judge was rearrested for a serious violent felony during pretrial release.

scribes the murder sentencing design. Section 3 describes the Philadelphia homicide docket design. Section 4 discusses mechanisms and potential institutional reforms. Section 5 concludes.

1 Data and Institutional Details

1.1 Cook County

Cook County is the largest county in Illinois, with five million residents, and the Circuit Court of Cook County is the second largest unified court system in the United States. The Cook County data come from the State’s Attorney’s Office, and contain the universe of adult felony defendants sentenced between January 2011 and November 2024 (Cook County State’s Attorney’s Office, 2024).⁷

The sample is composed of defendants who have either pled guilty or been found guilty in a bench or jury trial.⁸ Defendants may be sentenced to incarceration, or may be discharged. My primary outcome of interest is incarceration sentence length. 59% of defendants are incarcerated (nearly all non-incarceration sentences are probation sentences).⁹ Non-incarceration sentences are coded as zero-length, so the primary outcome includes all defendants.

Sentencing guidelines are pegged to the six felony classes, which in descending order of severity are Class M, X, 1, 2, 3, and 4. The nature of crimes across felony classes varies widely. For example, Class 4 felonies include domestic battery, obstruction of justice, and theft of more than \$500, and carry sentences between one and three years. Class X felonies include armed robbery, sexual assault, and home invasion, and carry sentences between six and thirty years. Judges have additional discretion in applying aggravating factors, such as the defendant’s criminal history or motivations, which may double the maximum sentence length. I cap sentence length at seventy-five years, assign life sentences this length, and top-code sentences at the relevant Illinois legal limit for their felony class. I observe both the initial felony class that prosecutors charge a defendant with, as well as the eventual felony charge class that a defendant is sentenced for, and control for both in all specifications.

The identifying variation comes from judges who sentence Class M felonies. The Class

⁷In 2017, State’s Attorney Kim Foxx began releasing data on the county’s handling of felony cases with historical records going back to 2011, but data release was discontinued after the election of State’s Attorney Eileen O’Neill Burke in 2024. The data are available [here](#).

⁸A diagram showing how cases move through the felony court system in Cook County is available [here](#).

⁹I define sentences at the Illinois Department of Corrections, Cook County Department of Corrections, Cook County Boot Camp, Cook County Impact Incarceration Program, and Juvenile IDOC as incarceration sentences. The vast majority of such sentences are served at the Illinois Department of Corrections and the Cook County Department of Corrections.

M felony charge is reserved for first-degree murders, which are murders that are intentional or are committed during the course of a felony. Class M felonies carry a sentence of twenty to sixty years, though defendants can face a life sentence if there are aggravating factors.

A desirable feature of the Cook County setting is that there is approximately random assignment of cases to judges within each courthouse (Loeffler, 2013). In an effort to preserve random assignment, I restrict the sample to judges who have sentenced at least 100 defendants, who have sentenced at least one Class M felony, and who are not assigned to problem-solving courts (e.g., domestic violence court).¹⁰

Additionally, the Cook County data provide information on the defendant’s race, gender, and age. I group age into five-year bins, with separate bins for all defendants below twenty or above sixty-five, in order to flexibly control for judge leniency patterns with respect to age which may be non-linear.

Following Abrams et al. (2012), I define an observation as a judge-defendant-calendar date tuple.¹¹ Importantly, while I observe defendant/co-defendant identifiers within a case across charges, I do not observe identifiers across cases, so I cannot control for criminal history. For each observation, I keep the longest sentence and most severe charge. This is a practical restriction, as empirically judges tend to record the exact same sentence when a defendant is convicted of multiple charges, even if those charges should carry different sentence lengths.¹² After imposing these sample restrictions, the sample contains 84 judges who sentence 145,412 cases including 1,184 first-degree murders.

1.2 Pennsylvania

Pennsylvania is the fifth largest U.S. state with a population of thirteen million. I study sentencing in the Pennsylvania Courts of Common Pleas, the state’s trial courts. This is a large court system: it is divided into sixty judicial districts and sentences over 70,000 defendants each year.

The Pennsylvania data come from the Pennsylvania Commission on Sentencing, a body tasked with the creation of sentencing guidelines and the maintenance of data on compliance

¹⁰There are three judge-level reasons why some judges may not be randomly assigned cases. First, some judges are floaters who pick up slack. Second, some judges are associate judges who need permission from the head judge to preside over felonies. In practice, many associate judges do preside over felonies and are given calls, but not all. Third, some judges are assigned to a problem-solving court. The first two reasons motivate the restrictions requiring sentencing of at least 100 defendants and at least one Class M felony, while the third motivates the exclusion of problem-solving court judges.

¹¹In an effort to be conservative, I also treat cases where there is a match on judge-date-demographics-sentence length as the same defendant, which back-of-the-envelope calculations suggest is true 92% of the time.

¹²Moreover, I do not observe whether multiple sentences are concurrent or consecutive, so I cannot combine sentences.

with those guidelines (Pennsylvania Commission on Sentencing, 2025). The data cover the near-universe of felony and misdemeanor sentences issued by the Pennsylvania Courts of Common Pleas, with some important exceptions: notably, for the research questions of this paper, the reporting of murder sentencing is encouraged, but not required, which may bias results towards zero.¹³ I study all years of data for which judge identifiers are available (1991–2024, excluding 1995).

In Pennsylvania, every sentence includes a minimum and maximum term. The minimum term is the amount of time until a defendant is eligible for parole. The maximum term is the amount of time until the state must release the defendant, and is legally required to be at least twice the minimum. My primary outcome of interest is the minimum sentence. Only 47% of defendants are sentenced to any incarceration.

Beyond the larger sample size, a desirable feature of the Pennsylvania setting is the availability of fine-grained information on charge characteristics and defendant history with respect to the sentencing guidelines. The sentencing commission recommends that judges sentence on the basis of the interaction between the Prior Record Score (PRS) and the Offense Gravity Score (OGS). The Prior Record Score includes nine levels ranging from no prior record to repeat violent offender.¹⁴ The Offense Gravity Score ranges from 1 to 15, with higher numbers reflecting more severe crimes. Within a given sentencing guideline era (seven editions of the guidelines are released over the course of the sample), a $\text{PRS} \times \text{OGS}$ cell fully captures the legally relevant information from the sentencing commission’s point of view regarding a suggested sentence range, though judges may apply sentencing enhancements. In all specifications, I flexibly control for sentencing guideline cells using $\text{PRS} \times \text{OGS} \times \text{Guideline Era}$ fixed effects.

The identifying variation comes from judges who sentence Grade 10 and 9 felonies, which correspond to first- and second-degree murders. Grade 10 felonies carry either life or death sentences, while Grade 9 felonies carry a life sentence.

In addition to sentencing information, the Pennsylvania data provide information on the defendant’s race, gender, and age. I group age into five-year bins, with separate bins for all defendants below twenty or above sixty-five.

As in Cook County, I define an observation as a judge-defendant-calendar date tuple. For each observation, I keep the longest sentence and highest charge. I do not observe defendant identifiers across cases, but because of the prior record score, I am able to control for criminal history. After imposing these sample restrictions, the sample contains 855

¹³See the Pennsylvania Commission on Sentencing documentation for more details on which cases are not covered; these include, but are not limited to, sentences from the Philadelphia Municipal Court and the magisterial district judges who typically handle DUIs and other misdemeanors.

¹⁴The levels are 0–6, RFEL, and REVOC, but which labels are available is era-dependent.

judges who sentence 2.5 million cases including 1,912 first- and second-degree murders in Pennsylvania (excluding Philadelphia) and 1,618 in Philadelphia.

1.3 Descriptive Statistics

Table 1 reports descriptive statistics for the Cook County, Pennsylvania, and Philadelphia samples. The Pennsylvania sample is more female and much Whiter than the Cook County sample, while the Philadelphia sample looks similar to Cook County in terms of gender and racial composition. Murder cases make up about 0.8% of the sample in Cook County and Philadelphia, but are ten times rarer in Pennsylvania. Incarceration rates are similar in Cook County and Philadelphia (58-59%), but lower in Pennsylvania (47%). Average sentences are roughly 2.6 years in Cook County. Average minimum sentences are 1.8 years in Philadelphia but around 5 months in Pennsylvania, reflecting its larger misdemeanor share.

1.4 Random Assignment

In this paper, I appeal to random assignment both to bolster identification arguments and to justify standard error choices. In Cook County, randomization is well-established (Abrams et al., 2012; Loeffler, 2013; Jordan et al., 2025). However, there are some known violations of random assignment such as high-profile cases, state benefits fraud, and probation violations which I cannot purge from the data.¹⁵ In Pennsylvania, Leibovitch (2016) finds evidence consistent with random assignment in most judicial districts. However, there are no institutional guarantees of random assignment, and in larger districts, there are some established violations such as problem-solving courts.¹⁶

Table 2 formally tests for random assignment of cases to judges in Cook County and Pennsylvania (excluding Philadelphia) conditional on courthouse by month and judicial district by month fixed effects, respectively. Cook County results are consistent with randomization: judge identity explains less than 0.6% of the variation in any defendant or case characteristic.

¹⁵According to Loeffler (2013), case assignment is handled by the presiding judge who uses a program called “the randomizer” to assign new cases to available judges in view of the Cook County State’s Attorney’s Office and Cook County Public Defender’s Office. However, based on a conversation I had with the authors of Jordan et al. (2025), there are several exceptions. First, some cases are siphoned off to problem-solving courts which are designed to handle sentencing related to mental health, drug addiction, and veterans’ issues. Second, if a defendant violates their probation, they return to the judge who originally sentenced them. Third, some state benefits fraud cases are not randomly assigned due to involvement by the governor’s office. Fourth, some high profile cases, colloquially referred to as “heaters,” are assigned to judges who are comfortable with increased attention and scrutiny. Fifth, some judges are floaters who do not have a permanent courtroom but instead pick up slack. Outside of these exceptions, randomization occurs at the “call” level, which represents a fixed courtroom.

¹⁶For example, Allegheny has Domestic Violence, DUI, Drug, Mental Health, Sex Offender, and Veterans courts.

In Pennsylvania, defendant demographics are similarly well balanced across judges within judicial district by month cells, but judge fixed effects explain 1.3% of the variation in felony status, which may reflect problem-solving court assignments or other forms of rotation. For example, while no other judicial district appears to have as strong homicide docket rotations as Philadelphia, some judges in Allegheny County appear to have cyclical exposure to higher and lower severity caseloads.

2 Murder Sentencing Design

2.1 Empirical Strategy

I want to recover the causal effect of sentencing a murder on other sentencing decisions. To do this, I estimate a modified stacked difference-in-differences design. Modifications are required because comparing a judge who sentences a murder to peer judges who do not departs from the canonical difference-in-differences settings in three ways. First, each judge can be treated multiple times. Second, the time between treatments varies, and the duration of treatment effects is unknown *ex ante*. Third, only a small number of judges sentence a murder on any given date. These differences preclude the straightforward application of estimators from the two-way fixed effects literature (e.g., Sun and Abraham (2021)) and recent stacked difference-in-differences estimators (e.g., Wing et al. (2024)) as these estimators rely on large cohorts being treated at the same time or assume that treatment is an absorbing state.

Instead, my design proceeds in two steps. First, I construct a stacked difference-in-differences panel by iterating through each murder sentence. A murder sentence is included as an event in the panel if the judge does not sentence a murder in the prior quarter, and if there is at least one month until the judge sentences another murder. The event window is defined to run from a quarter before the event until either a quarter after the event or the date of the next murder sentence, whichever occurs first. For each event, the control judges are the set of judges in the same courthouse (Cook County) or judicial district (Pennsylvania) who do not sentence a murder during the event window, and sentence at least one defendant on the day of the murder sentence. The stacked difference-in-differences panel is then constructed by including all sentences issued within each event window by treatment and control judges. Since the panel is constructed on an event-by-event basis, sentences may appear in the panel multiple times.¹⁷

¹⁷The inclusion rules bind: 363 of the 1,184 Cook County murder sentences and 1,065 of the 1,912 Pennsylvania murder sentences enter the estimation sample as events, and the resulting panels contain 79 of the 84 Cook County judges and 501 of the 689 Pennsylvania judges. The online appendix reports regression tables as well as plots of the component series underlying each of the difference-in-differences and triple differences

I purge unwanted variation from the panel by dropping the triggering murder sentences as well as co-defendants of the murderers. Because co-defendants present an important identification challenge, I take a conservative approach, dropping potential co-defendants anywhere within the event window. In Cook County, I drop explicit co-defendants who share a case ID with the murderer, but no such identifier is available in Pennsylvania. In both jurisdictions, I drop implicit co-defendants, which I define as defendants who share both the sentencing judge and the offense date of the murderer.

Second, using the panel, I estimate

$$Y_i = \delta \text{Treated}_{je} + \sum_{k \in \mathcal{W}} \beta_k \text{Treated}_{je} \cdot \mathbb{I}[k = w_i] + X_i' \gamma + \lambda_{w_i} + \psi_j + \alpha_e + \phi_c + \varepsilon_i, \quad (1)$$

where Y_i is sentence length in days of case i and Treated_{je} is an indicator for whether judge j was the treated judge in event e . $\mathcal{W}$ indexed by k is the set of event-time bins: these are weekly bins in relative time from thirteen weeks prior to the murder sentence until thirteen weeks after the murder sentence, and $w_i \in \mathcal{W}$ denotes the event-time bin in which case i is sentenced. Cases sentenced on the day of the murder sentence are assigned to a single-day bin indexed as event-time $k = 0$. The coefficients of interest are the treated event-time dummies β_k . The reference week is $k = -1$, the week prior to the murder sentence, and β_{-1} is normalized to zero. δ captures the reference-week treated-control difference. X_i are demographic controls (race, gender, age), λ_{w_i} are event-time fixed effects, ψ_j are judge fixed effects, α_e are event fixed effects, and ϕ_c are sentencing guideline fixed effects (initial felony class, ultimate felony class, and offense category in Cook County; Offense Gravity Score $\times$ Prior Record Score $\times$ Sentencing Guideline Era in Pennsylvania).

Standard errors are computed differently across jurisdictions to account for how case assignment works. In Cook County, the random assignment of cases to judges justifies independent and identically distributed (IID) standard errors as a baseline. Abadie et al. (2022) discuss when clustering is appropriate, and conclude in the context of an analogous design (judge leniency) that random assignment makes clustering unnecessary and observing the population of sentences makes robust standard errors excessively conservative.¹⁸

specifications.

¹⁸Quoting Abadie et al. (2022): “If one has a random sample of units from a large population with randomized treatment assignment at the unit level, there is no reason to cluster the standard errors of the least squares estimator... Similarly, in a judge-lenency design—where defendants are randomly assigned to judges—standard errors should not be clustered at the level of the judge (Chyn, Frandsen, and Leslie 2022). ... If the sample represents a large fraction of the population and treatment effects are heterogeneous across units, robust standard errors are also conservative. If the data contain information on attributes of the units that are correlated with unit-level treatment effects, the methods in Abadie et al. (2020) can be applied to obtain less conservative standard errors.”

In Pennsylvania, I do not report IID standard errors as a baseline, because the random assignment of cases to judges cannot be verified for each judicial district. However, because I observe the population of sentencing decisions, uncertainty in estimating treatment effects is design-based rather than sampling-based (Abadie et al., 2020). I estimate this uncertainty using randomization inference (RI) (Conley and Taber, 2011; MacKinnon and Webb, 2020).¹⁹ Section 2.5 discusses robustness to alternative methods of computing standard errors.

2.2 Identifying Assumptions

In order to recover the causal effect of sentencing a murder, the stacked difference-in-differences design requires two identifying assumptions. The first assumption is parallel trends: I assume that absent ruling on a murder, treated judges' excess sentencing would have evolved in parallel with that of control judges. The pre-periods of the panels in Figure 3 and Figure 5 are consistent with this assumption: conditional on observable controls, treated judges' excess sentencing behavior is similar in both level and trend to that of control judges.

The second identifying assumption is that the timing of treatment is not coincidental with something else that would change sentencing severity. In my view, the most serious version of this concern is the potential for sorting on unobservables around the time of the murder sentencing. For example, a judge could choose to sentence simpler cases while they handle the workload associated with more complex cases, or they might decide to sentence severe felonies in batches. These kinds of sorting would bias coefficients upwards, and would not be picked up by control judges, since control judges do not face the workload of a murder case by construction.

Figure 1 investigates the potential for sorting to drive the results in Cook County. I find evidence of sorting on observables: judges sentence high-severity cases 5 percentage points more often on the day of a murder sentence. However, conditional on ultimate felony class (which determines statutory sentencing ranges), I do not observe residual sorting on other observables. Each panel in Figure 1 plots difference-in-differences estimates. Panel A shows that treated judges sentence modestly more cases on the day of a murder sentence (0.4 more cases). Panel B emphasizes the main sorting concern: predicted sentence length on the basis of observables including felony class, offense category, and demographics is 81 days longer on the day of a murder sentence than the pre-period mean, though this effect is statistically

¹⁹Given that I observe the population of sentences, from a design-based perspective, the primary uncertainty is that I do not see what would have happened if a different judge had been treated (i.e., missing potential outcomes). Randomization inference estimates this uncertainty by randomly re-assigning the treated label among the treated judge and the control judges within each event and recomputing the estimator. Under the null of no effect, this procedure approximates the distribution of treatment effects. I take the standard deviation of the distribution as the standard error of the treatment effect.

insignificant. Panel C shows that this pattern is driven by judges sentencing 5 percentage points more high felony class cases on the day of a murder sentence. Note that this pattern cannot be driven by co-defendants who are already removed from the panel. Panel D shows that conditional on felony class there is no residual sorting. Given that the observed sorting is primarily about felony class composition changes, Panels E and F test whether the high felony class cases sentenced on the day of the murder sentence are observably more severe than their nearby counterparts. I do not find evidence for this concern: compared to nearby X/1 class felonies, the felonies sentenced on the day of the murder do not have longer cases (a proxy for case complexity) and do not have higher original charge classes.

I perform a similar analysis in Pennsylvania in Figure 2. In Panel A, I do not find evidence of a spike in volume on the day of the murder, but judges sentence 0.2 fewer cases per day in the two weeks prior to the murder. Panel B shows that predicted sentences rise on the day of the murder, and Panel C shows that this correlates with a 2 percentage point rise in sentencing felonies. Panel D shows that there is no predicted excess sentencing conditional on sentencing guideline cell. Panels E and F consider other proxies for case severity: I find no evidence of sorting on whether a case carries a mandatory minimum or the number of co-defendants.

The presence of sorting on observables makes sorting on unobservables an important identification concern. However, Figures 1 and 2 show that there is little evidence of residual sorting after conditioning on the information that determines sentencing ranges. Two further pieces of evidence support this conclusion. First, I compute Oster (2019) bounds in Table 3, and find that unobservable selection would need to be 14 times stronger than observable selection to drive treatment effects to zero. Second, there is no evidence of sorting on observables in the two weeks following a murder, yet I find treatment effects during these weeks for severe cases in Cook County and overall in Pennsylvania.

2.3 Cook County Results

Sentencing a murder causes judges to issue unusually long sentences to other defendants who happen to be sentenced on the same day. Panel A of Figure 3 reports coefficients from the main difference-in-differences design. Throughout the discussion of the murder design, I describe treatment effects relative to the pre-period mean.²⁰ On the day of a murder sentencing, sentences are 139 days longer, which is a 16% increase over the mean sentence

²⁰I say that the effect at event-time k is $\theta_k = \beta_k - \frac{1}{13} \sum_{j=-13}^{-1} \beta_j$, where the reference coefficient β_{-1} enters the sum as zero. This choice avoids over-emphasizing the level of the $k = -1$ reference bin, and tends to reduce the reported effect size. When reporting effects in percentage terms, I divide θ_k by the pre-period mean.

length of 885 days.

Panels B and C decompose the overall effect into intensive and extensive margins. Panel B restricts attention to incarceration sentences, and shows that those sentences are 243 days longer on the day of a murder sentencing. Panel C plots the probability of incarceration. Treatment effects are indistinguishable from zero. Together the results show that murder sentences cause judges to lengthen the sentences of defendants who would already face incarceration.

Figure 4 reports heterogeneous treatment effects using triple-differences. Panel A compares defendants sentenced for high felony class (X/1) charges to those sentenced for low felony class (2/3/4) charges. Effects are 531 days longer for high felony class cases on the day of a murder sentence, and remain elevated by 118 days for the following two weeks.²¹ However, the pre-period coefficients also show that high felony class sentences have large week-to-week variance, so this result should be interpreted with caution.

Panel B of Figure 4 shows that effects are 467 days longer for minority defendants than White defendants. Decomposing this effect, sentences rise by 185 days for minority defendants while the effect on White defendants is, if anything, negative (-282 days, imprecisely estimated). The effect is large: 185 days is a 20% increase above the mean sentence issued to a minority defendant, and is equivalent in size to the within-felony class minority-White sentencing gap (177 days).

2.4 Pennsylvania Results

Results are qualitatively similar in Pennsylvania. Because the Pennsylvania sample includes misdemeanors, sentences are much shorter than in Cook County: the average minimum sentence is 123 days (vs. 885 days) and the median minimum sentence is 0 days (vs. 365 days).

Panel A of Figure 5 reports coefficients from the main difference-in-differences design. On the day of a murder sentencing, sentences are 14 days longer, or 11% of the mean sentence length, a similar treatment effect size to Cook County in percentage terms. Sentences remain elevated in the two weeks following a murder.

Panels B and C of Figure 5 decompose the overall effect into intensive and extensive margins. Conditional on being incarcerated, sentences rise by 27 days on the day of a murder sentence and remain elevated for two weeks. As in Cook County, I cannot reject the null hypothesis that there is no effect on the probability of incarceration.

²¹When I report multi-week effects, these are simple averages of the weekly coefficients with standard errors computed following each jurisdiction's baseline method. When the day of the murder sentence is included, it is weighted by the average share of a week's cases sentenced on a murder day in that jurisdiction.

Panels A and B of Figure 6 report heterogeneous treatment effects using triple-differences. Panel A compares defendants charged with felonies and misdemeanors. Treatment effects are noisily estimated, but pooling the day of the murder with the following two weeks, I find that treatment effects are 20 days longer for felony defendants. As in Cook County, the high week-to-week variance in felony sentences means that this result should be interpreted with caution. Panel B estimates heterogeneity by race. While point estimates are directionally consistent with larger treatment effects on minority defendants, estimates are noisy and I cannot reject the null of no differential treatment effect.

In Panel C of Figure 6, treatment is interacted with the number of prior murders that a judge has sentenced. For each additional murder, the over-sentencing penalty rises by 4 days, which is roughly a quarter of the overall treatment effect of 14 days.²²

2.5 Robustness

Figure 7 reports a battery of robustness tests for Cook County and Pennsylvania with respect to alternative standard errors, window length, additional controls, and alternative weighting schemes. I report results under four methods of computing standard errors that are increasingly conservative: IID, RI, causal cluster variance (CCV) (Abadie et al., 2022), and event-level clusters. I consider robustness to window length, which changes the requirements for murder sentencing event inclusion. Longer windows impose the requirement that a judge not sentence a murder for a longer pre-period: this excludes more events, but the surviving events are less contaminated by recent murder exposure. The baseline specification uses a one quarter (thirteen-week) window, but I test robustness to six- and nineteen-week windows. Additional controls include case length (a proxy for complexity) and daily caseload in both samples, plus indicators for mandatory minimum application and trial disposition in Pennsylvania. Finally, I implement Q-weighting as prescribed by the Wing et al. (2024) stacked difference-in-differences estimator, which equalizes the weight of the control cases to the treatment case within each panel.

For each jurisdiction, I report the robustness of three headline results: overall treatment effects, charge severity triple differences, and minority-White triple differences. For each result, I report effects on the day of the murder sentence, and test for persistence in the two weeks following the murder.

In Cook County, results on the day of the murder are broadly robust, although only the minority-White difference survives the more conservative standard errors (CCV, event

²²I do not report this result in the Cook County sample because the panel is short, which causes measurement issues: I do not know the full history of murder sentencing for the majority of judges.

clustering). Evidence for persistence is limited to high-severity cases, and this result is sensitive to window length and standard errors.

In Pennsylvania, overall treatment effects are broadly robust, both on the day of the sentencing as well as in the two weeks that follow (however, neither survives CCV or event clustering). The charge severity difference is more noisily estimated: effects on the day of the murder are sensitive, but effects over the subsequent two weeks are broadly robust (though, again, these do not survive more conservative standard errors). I largely do not find evidence of differential effects by race across specifications.

Figure 7 also reports effects on the maximum sentence in Pennsylvania. In general, I cannot reject null effects. Given the effects on minimum sentences, this is surprising, but it stems from two facts. First, while point estimates tend to be similar to minimum sentence treatment effects, standard errors are twice as large, reflecting the larger variance in maximum sentences. Second, the maximum sentence is affected by ceiling effects: roughly a quarter of incarcerated sentences are assigned the statutory maximum within a sentencing guideline cell.

3 Philadelphia Homicide Docket Design

3.1 Institutional Context

The murder design estimates the causal effect of sentencing a single case, but how does sustained exposure to extreme cases affect judicial decisions? I leverage a quirk of the Philadelphia judicial district of the Pennsylvania Courts of Common Pleas to study this question. In Philadelphia, the supervising judge assigns other judges to various dockets including homicide, major felonies, and mental health. Because of this case assignment mechanism, when judges rotate onto the homicide docket, exposure to murder cases sharply rises and persists, typically for multiple years. I use this natural variation to estimate the causal treatment effect of rotating onto and off of the homicide docket on the sentencing of non-murder defendants.

3.2 Empirical Strategy

I estimate homicide docket rotations empirically, because the exact dates of when judges rotate onto and off of the homicide docket are not publicly documented. I say that a judge is on homicide docket rotation if they sentence a murder on at least four distinct dates where each consecutive pair of dates is no more than one year apart. If a year elapses between clusters of murder sentencing dates, I treat these clusters as separate rotations. I define the

start of the rotation using the date of the first murder sentence within the cluster, and the end using the date of the last murder sentence within the cluster. Figure 8 provides a judge-by-judge visualization of the identifying variation. The Philadelphia sample is composed of 166 judges who sentence 193,069 cases. Using this method, I detect 59 homicide rotations across 39 judges, and these judges sentence 67,450 cases.

Figure 9 compares the distribution of grades, offense gravity scores, and prior record scores between homicide docket judges while on rotation and never-rotated judges over the same periods of time. While judges on homicide docket rotation sentence more severe cases on average, there is common support across the distribution of each measure. Even within bins that appear visually unbalanced such as OGS 14 (which includes third-degree murder among other serious crimes), never-rotated judges outnumber rotator judges, so the total number of OGS 14 cases sentenced by treated and control judges is comparable (907 vs. 718). The one exception to common support is first- and second-degree murders (grade 9/10 cases). 98% of all first- and second-degree murder cases in Philadelphia are handled by judges who rotate onto the homicide docket, and 96% fall within rotation windows. For this reason, I restrict attention to the sentencing of non-murder cases. As a precaution, I also drop non-murder co-defendants to murder cases from the analysis.

With rotations identified, I estimate the stacked difference-in-differences design in three steps. First, to preserve comparability in levels between the rotate-on and rotate-off panels, I residualize sentence length by estimating

$$Y_{ijc} = X_i' \beta + \alpha_j + \gamma_c + \varepsilon_i,$$

where Y_{ijc} is sentence length in days of case i sentenced by judge j under sentencing guideline cell c , X_i are demographic controls (race, gender, age), α_j are judge fixed effects, and γ_c are sentencing guideline-cell fixed effects. I define excess sentence length as the residual given by $\tilde{Y}_i := Y_{ijc} - X_i' \hat{\beta} - \hat{\alpha}_j - \hat{\gamma}_c$.

Second, I iterate through rotations to construct stacked difference-in-differences panels. Each rotation contributes to the rotate-on design if there is at least one quarter of observed pre-period, and contributes to the rotate-off design if there is at least one quarter of observed post-period (53 and 48 of the 59 rotations, respectively). For each rotation, I construct a set of control cases composed of all cases sentenced by never-rotating judges in Philadelphia during the same time period. These panels are unbalanced: rotations vary in length, as do the observed pre- and post-periods. I study up to three years before or after each rotation.

Third, I estimate the following equation for the rotate-on and rotate-off panels separately:

$$\tilde{Y}_i = \sum_{q \in \mathcal{Q}} (\beta_{0,q} + \beta_{1,q} \mathbb{I}[\text{Homicide Docket Judge}]) \mathbb{I}[q_i = q] + \varepsilon_i,$$

where q indexes event time in quarters, the $\beta_{0,q}$ are common event-time levels absorbed as quarter fixed effects, and the coefficients of interest, $\beta_{1,q}$, are the within-quarter differences in excess sentence length between homicide docket judges and control judges, estimated freely in every quarter.

As in Pennsylvania, I report design-based standard errors computed by randomization inference as a baseline. Since I observe the universe of misdemeanor and felony sentences, the design uncertainty derives from the realization of treatment assignment (i.e., which judges rotate onto the homicide docket). With only 59 rotations, randomization inference is a natural choice because it does not rely on cluster asymptotics (Conley and Taber, 2011; MacKinnon and Webb, 2020).²³ Section 3.5 discusses robustness to alternative methods of computing standard errors.

3.3 Identifying Assumptions

In order to identify the causal effect of rotating onto or off of the homicide docket on sentencing decisions, this design requires two assumptions. The first assumption is parallel trends: if judges who rotated onto or off of the homicide docket had not done so, I assume that their sentencing behavior would have evolved along the same path as never-rotators. The pre-periods of the rotate-on and rotate-off panels in Figure 10 appear largely consistent with this assumption. One potential concern is that sentences in the two quarters prior to treatment appear elevated, albeit insignificantly. One plausible explanation is that the estimate of the date of rotating onto the homicide docket is conservative, since the rotation starts on the day of the first murder sentencing, but judges handle the murder case for some time prior to sentencing.²⁴

The second assumption is that rotating onto or off of the homicide docket is not coincidental with something that would only affect the sentencing behavior of treatment judges. One version of this concern is that rotating onto the homicide docket could coincide with something else in a judge’s life that affects sentencing. Since assignment to the homicide

²³Within each rotation window, I randomly re-assign the rotator label among the rotating judge and the never-rotated judges active in the same period and recompute the estimator. Under the null of no effect, this procedure approximates the distribution of treatment effects. I take the standard deviation across re-assignments as the standard error.

²⁴In Philadelphia, one judge handles all pre-trial hearings for homicide cases, but treated judges may handle trials before sentencing.

docket is determined by the supervising judge and therefore is not random, I cannot definitively reject this story, but the sharp timing of the rotate-on treatment effects pushes against it.

In my view, the most plausible threat to the second assumption is the possibility that the mix of cases that judges see when they rotate onto the homicide docket is more severe in an unobservable way. Sorting on unobservables is plausible since rotating onto the homicide docket guarantees that judges will preside over an observably more severe mix of cases. From a sentencing equity perspective, the guidelines-cell fixed effects should resolve the bulk of this concern. In particular, the Offense Gravity Score ensures that I compare charges of legally equivalent severity, and the Prior Record Score ensures that I compare defendants with legally equivalent prior records. So, even if cases sentenced while on the homicide docket are unobservably different, they are not unobservably different in ways that cross narrow sentencing-guideline thresholds. The distributions of charge severity and defendant history reported in Figure 9 are reassuring: common support along the entire severity spectrum pushes against the idea that homicide docket judges are assigned entirely non-comparable caseloads. To further assuage this concern, I compute Oster (2019) bounds in Table 3, and find that unobservable selection would need to be at least 10 times stronger than observable selection to drive the overall rotate-on and rotate-off treatment effects to zero.

3.4 Results

Philadelphia judges see relatively more severe cases than the average Pennsylvania judge. Among the set of non-murder cases that I study, the mean minimum sentence is 617 days and the median minimum sentence is 183 days.

Panels A and B of Figure 10 plot the overall effect of rotating onto and off of the homicide docket. Starting from the first quarter after the rotation begins, there is a sharp rise in sentence length. Throughout the discussion of the homicide design, I summarize treatment effects as the difference between the excess sentencing in the pre- and post-periods.²⁵ While on the homicide docket, sentences are 125 days longer, which represents a sustained 20% increase in sentence length. After rotating off of the homicide docket, sentences do not immediately fall, but appear to remain elevated for at least three months after rotation, before returning towards the baseline. Comparing sentencing during the homicide docket to the three years after rotating off of the docket, excess sentence length declines from 125 days to 45 days on average, reversing 64% of the rotate-on treatment effect.

Panels C–F of Figure 10 decompose the main treatment effect into intensive and ex-

²⁵Specifically, I say that the effect of a rotation is $\theta = \frac{1}{12} \sum_{q=1}^{12} \beta_{1,q} - \frac{1}{12} \sum_{q=-12}^{-1} \beta_{1,q}$.

tensive margins. Panels C and D restrict attention to cases with positive sentence length. Conditional on incarceration, sentences rise by 143 days after rotating onto the homicide docket, and fall by 109 days after rotating off, though, as with the main treatment effect, the fall does not begin until three months after rotation. Panels E and F estimate effects of the homicide docket on incarceration probability. As in Cook County and Pennsylvania, I cannot reject the null hypothesis that there is no effect on the probability of incarceration.

Figure 11 reports heterogeneous treatment effects using triple-differences. Panels A and B compare felony and misdemeanor sentencing: rotate-on effects are 186 days larger for felonies, and rotating off reverses 124 days or 67% of the differential treatment effect. Panels C and D compare minority and White defendants: rotate-on effects are 84 days larger for minority defendants. Unlike the overall and charge severity results, this differential effect does not appear as a sharp break at the rotation date, but rather accumulates over time. This means that the credibility of the differential race effects rests more heavily on guideline cell fixed effects, so results should be interpreted with caution. Differential rotate-off effects by race are imprecisely estimated, but point estimates imply that rotating off reverses 41% of the differential rotate-on effect for minority defendants.

3.5 Robustness

Figure 12 presents a similar set of robustness tests to those discussed in Section 2.5, but reports a different test statistic: the mean difference in excess sentence length between the pre- and post-periods.

Overall effects and differential effects by charge severity are broadly robust across alternative standard errors, controls, and weightings. Differential effects by race are more sensitive. Rotate-on effects do not survive more conservative standard errors (CCV, event-level clusters) or additional controls. I cannot reject the null of equal rotate-off treatment effects for minorities and White defendants across all specifications.

Figure 12 also reports effects of the homicide docket rotation on the maximum sentence. Unlike the Pennsylvania murder design results, I find effects on maximum sentence that are consistent in direction, equal or larger in magnitude than effects on the minimum sentence, and estimated precisely enough to be detectable. The Philadelphia effects are easier to detect because they are larger, and the rotation statistic averages sentences over a three-year treatment window rather than two weeks, which lowers its sampling variance.

4 Discussion

4.1 Mechanisms

As shorthand, I describe the spillover effect identified in this paper as an assimilation effect, but the murder sentencing should be understood as a bundle of at least three treatments: (1) handing out numerically longer sentences, which may cause anchoring, (2) receiving new information, which may cause (potentially rational) updating to beliefs about the consequences of crime or the returns to incarceration, and (3) handling a complex, violent case, which may increase cognitive load, stir up emotions, or otherwise alter a judge’s mental state. Beyond these channels, the homicide docket rotation treatment also shifts the full case distribution.

The mechanisms behind the extreme case spillover effects I document are inherently difficult to disentangle. I do not observe detailed information about the timing of events within each case, or the behavior of prosecutors or defense attorneys, so I do not attempt to provide a definitive interpretation of what drives the effects. Despite these challenges, the results do provide some suggestive evidence.

Salience appears to mediate the spillover bias, consistent with the model proposed in Bordalo et al. (2015). The salience interpretation is supported by the heterogeneous treatment effects by race and charge severity, which are dimensions along which defendants resemble the typical murder defendant. Additionally, the fact that effects fade away after a murder sentencing/homicide docket rotation is consistent with a memory-based mechanism.

The results do not seem consistent with classic behavioral stories like anchoring. First, these stories have trouble explaining why spillover effects would persist for months after a homicide docket rotation ends. Second, behavioral biases typically shrink with experience and skill, as in Chen et al. (2016) and Angelova et al. (2025), but instead I observe spillover effects which persist for years during a docket rotation in Philadelphia and grow with the number of prior murders sentenced in Pennsylvania. The sustained nature of the homicide docket effects also pushes against emotion-based mechanisms, because it would be unusual for a heightened emotional state to last for months following the end of the rotation.

The results also do not seem consistent with standard learning or durable preference changes. While effects increase with experience in Pennsylvania, these same effects fade within weeks of a murder. Memory mechanisms could help explain this pattern: for example, triggering events could cause experienced judges to temporarily recall a larger set of related memories, resulting in a larger bias that still fades. Taking stock, the evidence appears to be most consistent with a memory mechanism in which salient cases distort subsequent sentencing.

4.2 Policy Relevance

Extreme case spillover effects are particularly policy-relevant because they manifest as a function of institutional design choices. It is within the purview of the court to decide how to assign cases to judges, which is clear from the fact that the Philadelphia judicial district assigns cases in a different way than the rest of Pennsylvania. While I do not take a stance on what case assignment mechanism would be welfare-maximizing, or whether the harsher or less harsh sentences are “correct,” it is clear that the spillover between cases violates the foundational legal principle of equal justice under the law.

This violation is worth correcting given the size, scope, and differential impact of the bias. The effect sizes I find are on the higher end of the spectrum within the literature on biases in judicial decisions. Sentencing a murder increases sentence length by 11–16% on the day of the sentence, and presiding over the homicide docket increases sentence length by 20% on an ongoing basis. As points of comparison, Cohen and Yang (2019) find that the federal Republican-Democratic sentence length gap is 11% (61.8 months vs. 55.5 months), and Philippe and Ouss (2018) find that coverage of crime on a national television newscast watched by 20% of households in France increases sentence length by 2.5% (three months on a base of ten years).

Another reason these results are concerning is their scope: judges sentence murders regularly, so spillovers affect a meaningful share of cases on an ongoing basis. After sentencing a murder, sentences remain elevated for two weeks in Pennsylvania and for two weeks for severe cases in Cook County. Empirically, 10% of such cases occur within two weeks of a murder in Cook County and 2% of such cases occur within two weeks of a murder in Pennsylvania. Similarly, the effects of homicide docket rotation persist for the entirety of the rotation and for up to three months afterward. In Philadelphia, 19% of cases sentenced by homicide docket judges fall within this range, and together these cases account for 7% of all cases sentenced in Philadelphia.

Additionally, the results present equity concerns: spillover effects are larger for minorities. In Cook County, the treatment effect on minorities is 185 days, while the effect on White defendants is, if anything, negative. In Philadelphia, rotate-on treatment effects are 2.4 times larger for minority defendants. These differential effects are large in the context of existing racial sentencing penalties: the gap between minority and White sentencing in Cook County conditional on felony class is 177 days.

The results suggest two potential policy interventions which could reduce the scope of the bias. First, judges who sentence severe cases could take cool-off periods: in Cook County and Pennsylvania, abstaining from sentencing other cases on the day of a murder sentence and taking two weeks off from sentencing after a murder sentence would mitigate the bias. The

main problem with this solution is inefficiency: a cool-off period would only be reasonable if judges could easily substitute between sentencing and other tasks.

A second solution would be to more fully segregate dockets by case severity. One reason the Philadelphia homicide docket generates bias is that the judges sentence both murders and a variety of other cases that range in severity. The spillover bias stems primarily from sentencing cases close in severity to murders. Forcing judges to only handle homicides while on the docket would resolve this issue, but would likely be inefficient. One alternative could be to have homicide docket judges handle only homicides and minor offenses. This proposal connects to the broader theory that assimilation effects tend to occur when cases are similar, while contrast effects tend to occur when cases are different, so dividing a judge’s docket into two distinct streams could reduce the potential for assimilation, but might increase the risk of contrast effects (Mussweiler, 2003). Given the narrow sentencing range of minor offenses, this could be a worthwhile tradeoff.

5 Conclusion

I document how extreme cases produce a sizable, ongoing bias in criminal sentencing in Cook County and Pennsylvania. On the day of a murder sentence, sentence lengths for unrelated defendants increase by 11–16%. Effects persist for two weeks in Pennsylvania and for two weeks among severe cases in Cook County. When judges in Philadelphia rotate onto the homicide docket, sentence lengths for unrelated defendants rise by 20%; sentence lengths fall by 13% when judges rotate off. Homicide docket effects persist for the duration of the rotation and three months afterward. This spillover bias exacerbates existing racial disparities: effects in Cook County temporarily double the minority-White sentencing gap, and effects in Philadelphia are 2.4 times larger for minority defendants. Taken together, these results document a violation of the equal justice principle in which defendants’ sentences depend on the luck of timing.

From a behavioral economics perspective, these results contribute to the growing literature on assimilation and contrast effects in the field by showing how extreme cases can create durable assimilation effects lasting weeks to months after exposure ends. The judges I study are experienced decision-makers facing high stakes, yet I find effects that are increasing in experience (in Pennsylvania) and that sustain for years-long rotations (in Philadelphia). These patterns, together with the fact that effects fade over time and tend to be larger for high-severity crimes and minority defendants, are suggestive of a memory mechanism in which salient cases distort subsequent sentencing.

From a legal policy perspective, the results highlight how behavioral biases can inter-

act with institutional design choices to produce unequal treatment. The comparison of Philadelphia to the rest of Pennsylvania demonstrates how case assignment mechanisms directly influence the size and prevalence of the spillover bias. The biases I uncover could be remediated by solutions that are within the purview of the court to implement: introducing cool-off periods or more fully segregating dockets by case severity could reduce this form of sentencing bias. More generally, these results suggest that even when expertise and high stakes fail to discipline behavioral biases, institutional design that accounts for such bias can limit its consequences.

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Table 1: Descriptive Statistics by Jurisdiction

	Cook County	Pennsylvania	Philadelphia
Demographics			
Female	10.9	21.1	11.3
Age	32.5 (11.7)	33.9 (11.2)	31.5 (10.5)
<i>Race</i>			
White	13.0	69.7	20.3
Black	69.1	24.6	63.8
Hispanic	16.7	2.6	7.9
Asian	0.5	0.3	1.3
American Indian	0.0	0.1	0.0
Other	0.0	0.1	0.5
Missing	0.6	2.5	6.2
Case Characteristics			
Murder Case	0.81	0.08	0.84
Felony	–	27.8	76.1
Misdemeanor	–	72.1	23.1
Felony Class X	5.6	–	–
Felony Class 1	10.0	–	–
Felony Class 2	22.9	–	–
Felony Class 3	16.8	–	–
Felony Class 4	43.8	–	–
Offense Gravity Score	–	3.6 (2.4)	6.0 (2.8)
Prior Record Score	–	1.3 (1.8)	1.6 (2.0)
Repeat Felony Offender	–	2.6	3.3
Repeat Violent Offender	–	0.0	0.3
Mandatory Minimum	–	24.7	5.1
Sentence Enhancement	–	1.3	2.9
Convicted at Trial	–	2.1	19.6
Case Length	315 (958)	352 (367)	490 (565)
Sentencing Outcomes			
Length	958.2 (2027.7)	154.7 (888.1)	652.0 (2608.4)
Length Incarceration	1621.0 (2425.2)	326.1 (1267.4)	1130.3 (3354.7)
Incarceration	59.1	47.4	57.7
Cases	145,412	2,261,659	193,069
Judges	84	689	166
Years Covered	2011–2024	1991–2024 (excl. 1995)	1991–2024 (excl. 1995)

Note: The Pennsylvania column excludes Philadelphia. Standard deviations are in parentheses. All variables are percentages except age (years), case length and sentence lengths (days), Offense Gravity Score (1–15), and Prior Record Score (0–6).

Table 2: Randomization Tests

	Cook County		Pennsylvania	
	F	ΔR^2	F	ΔR^2
Female	3.03	0.0016	6.88	0.0021
Black	13.03	0.0057	15.35	0.0042
Hispanic	3.12	0.0016	17.48	0.0047
Age	4.41	0.0023	9.75	0.0029
High Severity	6.16	0.0032	46.17	0.0134
Assignment pool	courthouse $\times$ month		judicial district $\times$ month	
Cases	145,412		2,261,643	
Judges	84		689	

Note: Each row reports regressions of the listed case characteristic on judge fixed effects, conditional on assignment-pool by time fixed effects (Cook County: courthouse $\times$ month; Pennsylvania: judicial district $\times$ month). F is the joint F-statistic on the judge fixed effects and ΔR^2 is the increase in R^2 from adding them. Pennsylvania excludes Philadelphia. High Severity is disposition class M, X, or 1 in Cook County and offense grade 5 or higher (felonies) in Pennsylvania. In Cook County, District 1's narcotics call and District 5's Branch 38 intake are treated as distinct courthouses.

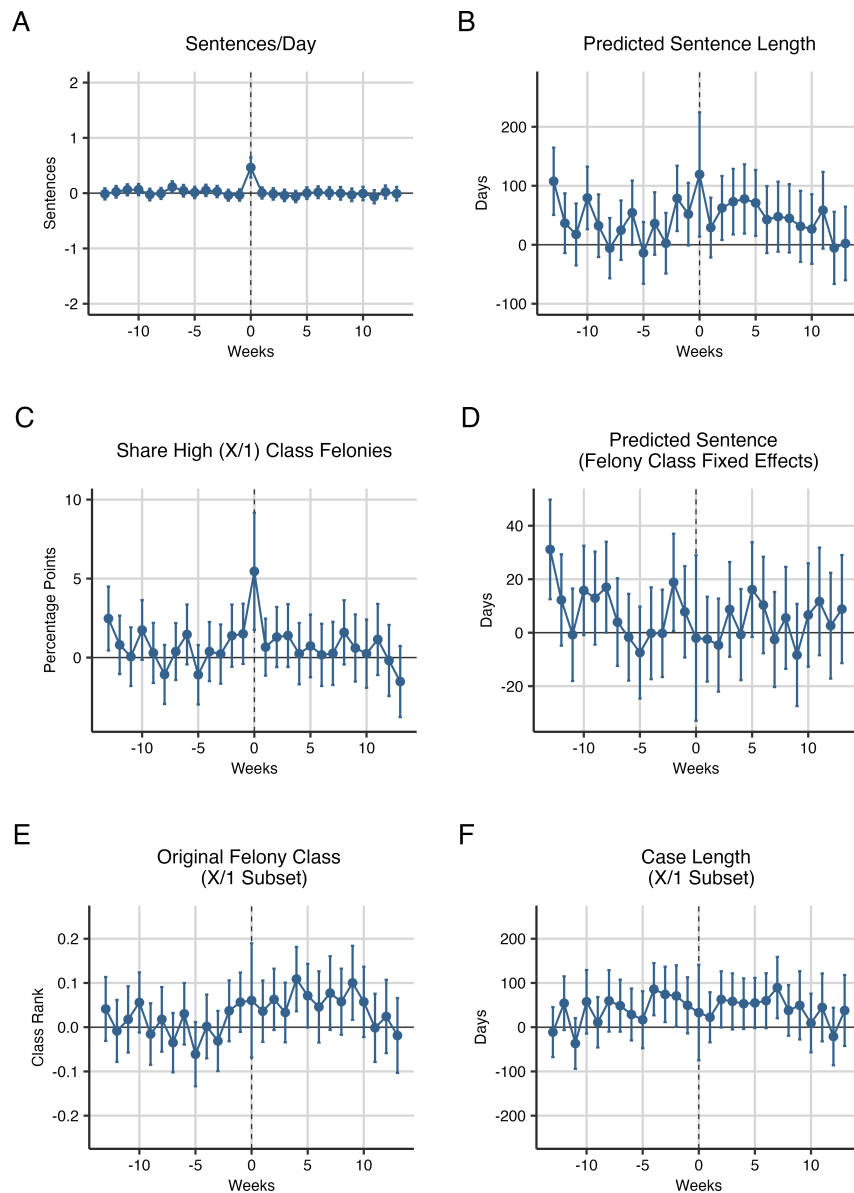
Table 3: Coefficient Stability Analysis

	Cook County	Pennsylvania	Philadelphia	
			Rotate On	Rotate Off
Baseline (β°)	161.6 (52.8)	14.4 (5.3)	126.7 (15.6)	-84.1 (17.9)
Full ($\tilde{\beta}$)	131.4 (51.4)	12.3 (5.0)	96.4 (14.5)	-64.1 (16.8)
Within- R^2 , Baseline ($R^{\circ 2}$)	0.000	0.000	0.000	0.000
Within- R^2 , Full ($\tilde{R}^2$)	0.051	0.110	0.107	0.107
$R^2_{\max} = 1.3\tilde{R}^2$	0.067	0.143	0.139	0.139
δ^* (selection ratio)	14.4	19.2	10.6	10.7
Statistic	Day 0 Effect		Pre-Post Gap	
Observations	215,680	1,081,536	956,110	798,143

Note: This table reports Oster (2019) selection-on-observables diagnostics. Each column compares baseline and full controls specifications. In Cook County, the baseline includes event, judge, initial/final class, and offense-category fixed effects. The controls specification adds demographics, case length and caseload. In Pennsylvania, the baseline specification includes event, judge, and guidelines-cell fixed effects. The controls specification adds demographics, case length, caseload, and flags for mandatory-minimums and trial disposition. Philadelphia matches Pennsylvania’s specification without event fixed effects. δ^* is the degree of proportional selection on unobservables, relative to selection on observables, required to drive the controlled effect to zero.

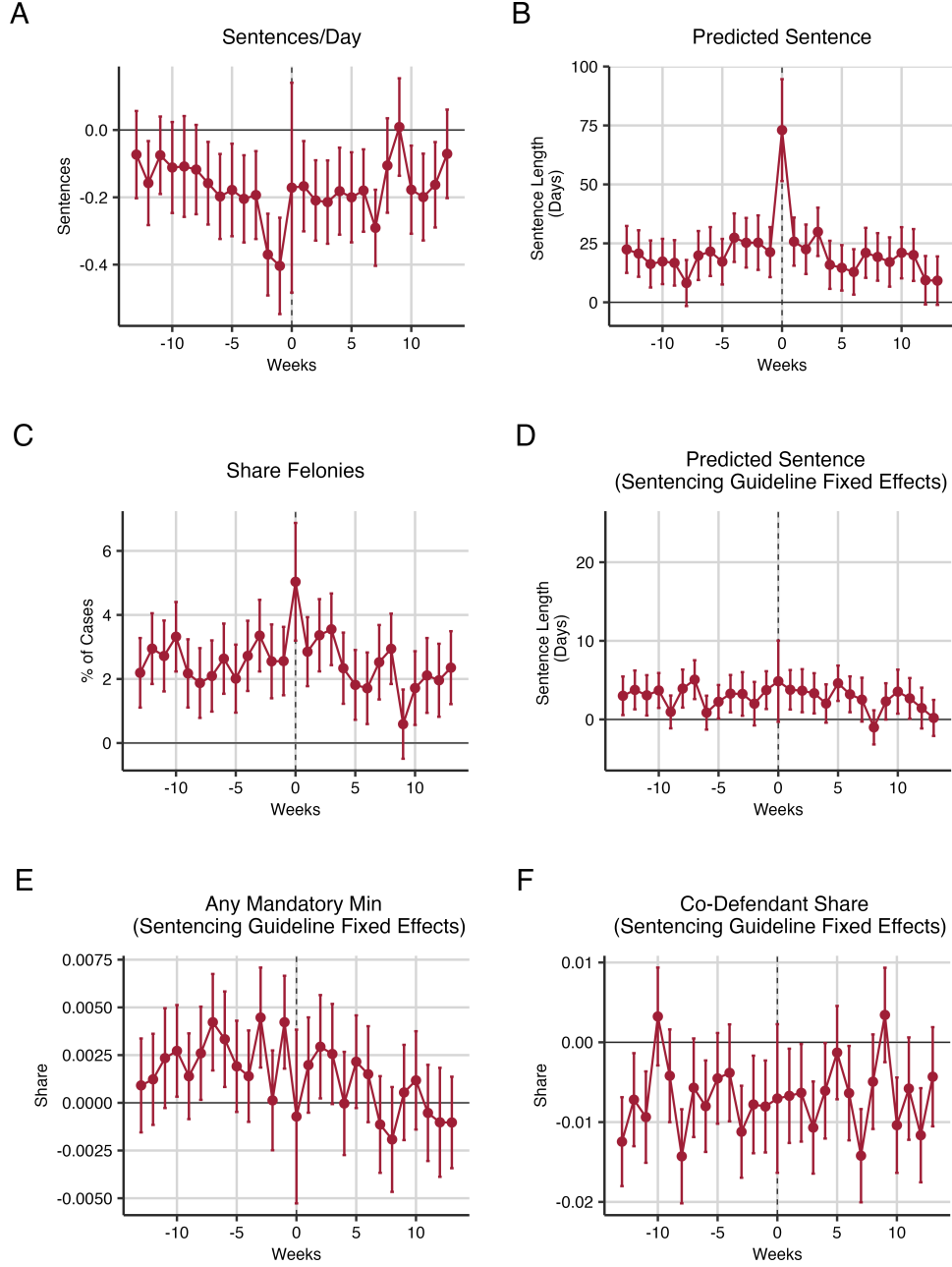
Figures

Figure 1: Sorting on Observables (Cook County)



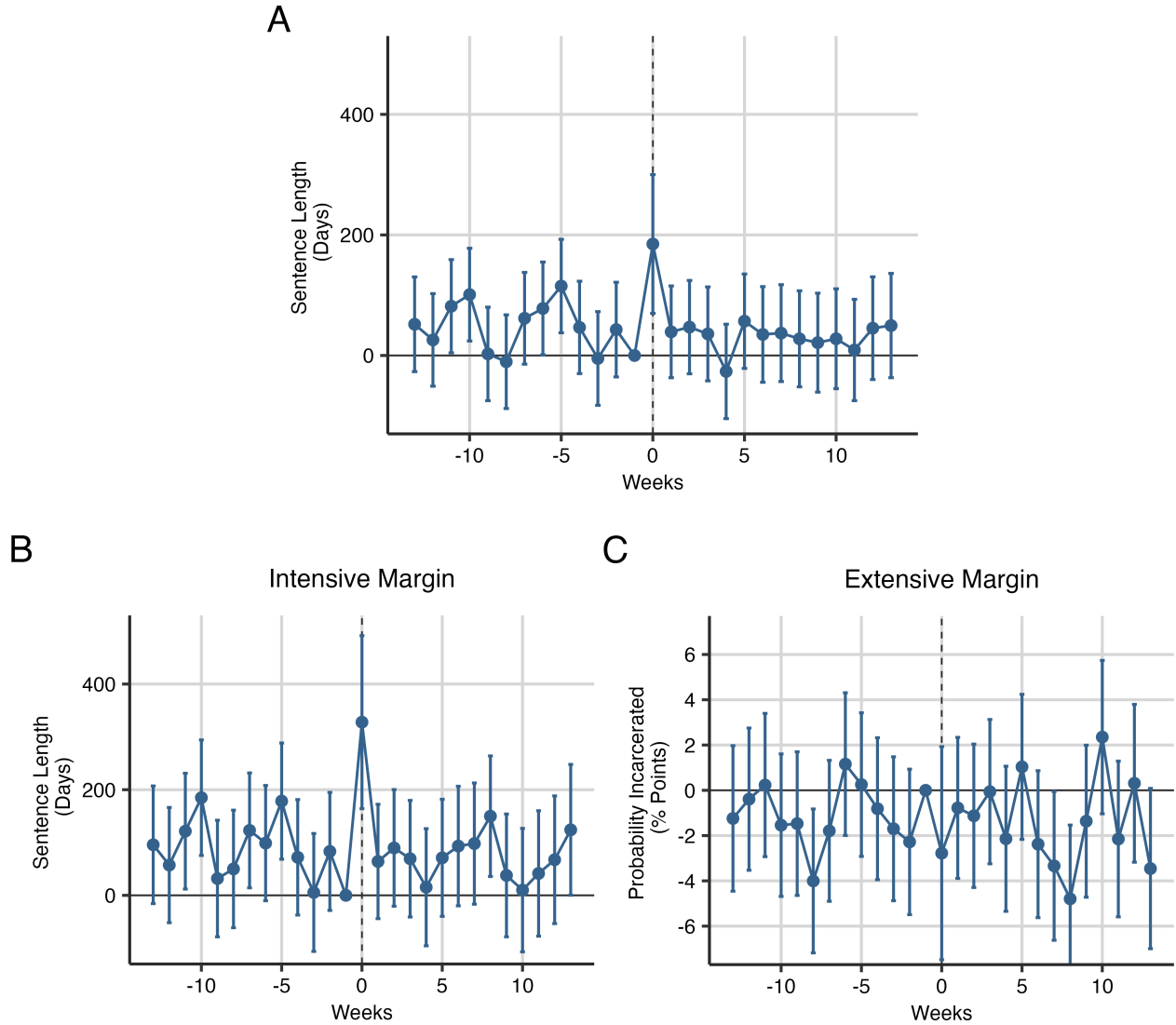
Note: The goal of this figure is to investigate the extent to which the treatment effects documented in Figure 3 may be driven by judges sentencing more severe cases at time 0. Each panel plots weekly treated-minus-control differences in means, comparing judges who sentence a murder to control judges from the same courthouse. Time 0 is the day of the murder sentencing. All other points represent weeks in relative time. Bars are 95% confidence intervals. Panels A–D document that judges sentence more severe cases at time 0, and that this is driven by sentencing more high-class felonies. Panels E and F show that high-class felonies sentenced at time 0 are similar on observables to other high-class felonies sentenced within the event window. Panels B and D plot predicted sentence length (fitted values from observable case and defendant characteristics). Panel D demeans predictions within final felony class. Panels E and F restrict to Class X/1 felonies.

Figure 2: Sorting on Observables (Pennsylvania)



Note: The goal of this figure is to investigate the extent to which the treatment effects documented in Figure 5 may be driven by judges sentencing more severe cases at time 0. Each panel plots weekly treated-minus-control differences in means, comparing judges who sentence a murder to control judges from the same judicial district. Time 0 is the day of the murder sentencing. All other points represent weeks in relative time. Bars are 95% confidence intervals. Panels A–D document that judges sentence more severe cases at time 0, and that this is driven by sentencing defendants who face more severe charges and/or have longer records. Panels E and F show that conditional on observable case severity, defendants sentenced at time 0 are similar on observables to other defendants sentenced within the event window. Panels B and D plot predicted sentence length (fitted values from observable case and defendant characteristics), and Panels D–F demean each outcome within sentencing guideline cells ($OGS \times PRS \times Era$) before differencing.

Figure 3: Average Treatment Effects and Margin Decomposition (Cook County)

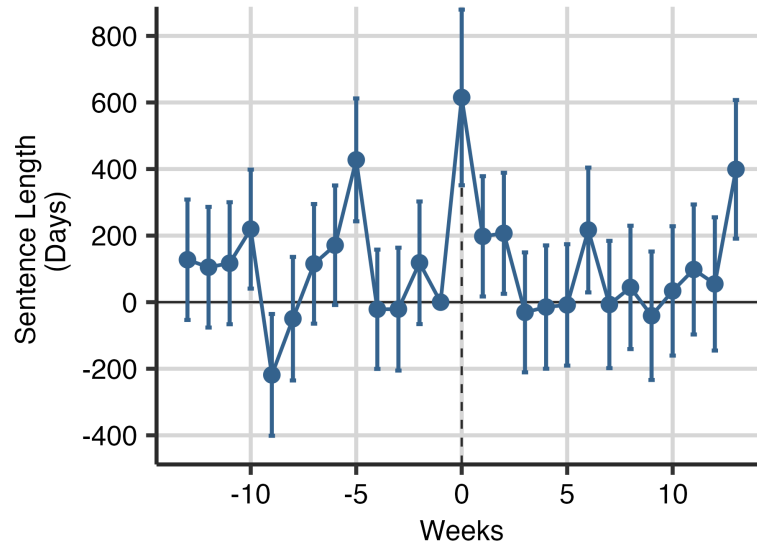


Note: Each panel in this figure plots coefficients from a difference-in-differences design comparing judges who sentence a murder to control judges from the same courthouse. Time 0 is the day of the murder sentencing. All other points represent weeks in relative time. Bars are 95% confidence intervals. Panel A plots the average treatment effects on sentence length. Panel B plots sentence length conditional on incarceration, and panel C plots incarceration probability. Regressions include controls for the initial felony charge class, the final felony charge class, the defendant's race, gender, and age, and fixed effects for the sentencing judge and the murder sentencing event.

Figure 4: Heterogeneous Treatment Effects (Cook County)

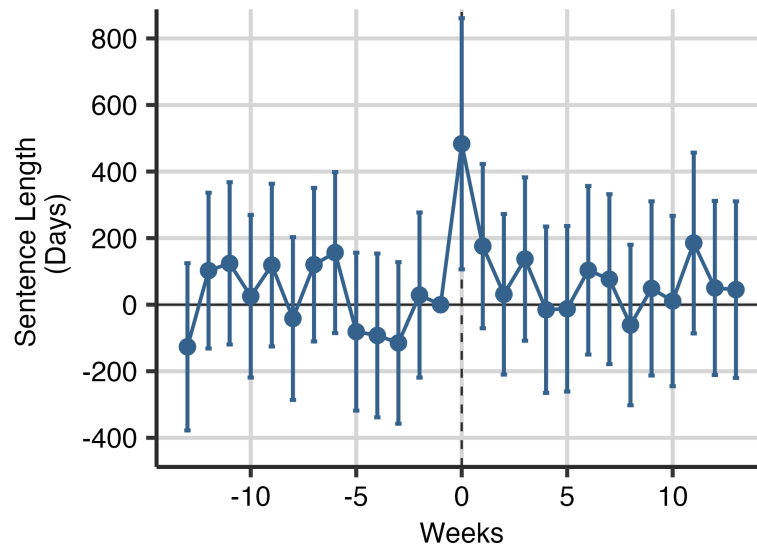
A

High Felony Class (X/1) – Low Felony Class (2/3/4)



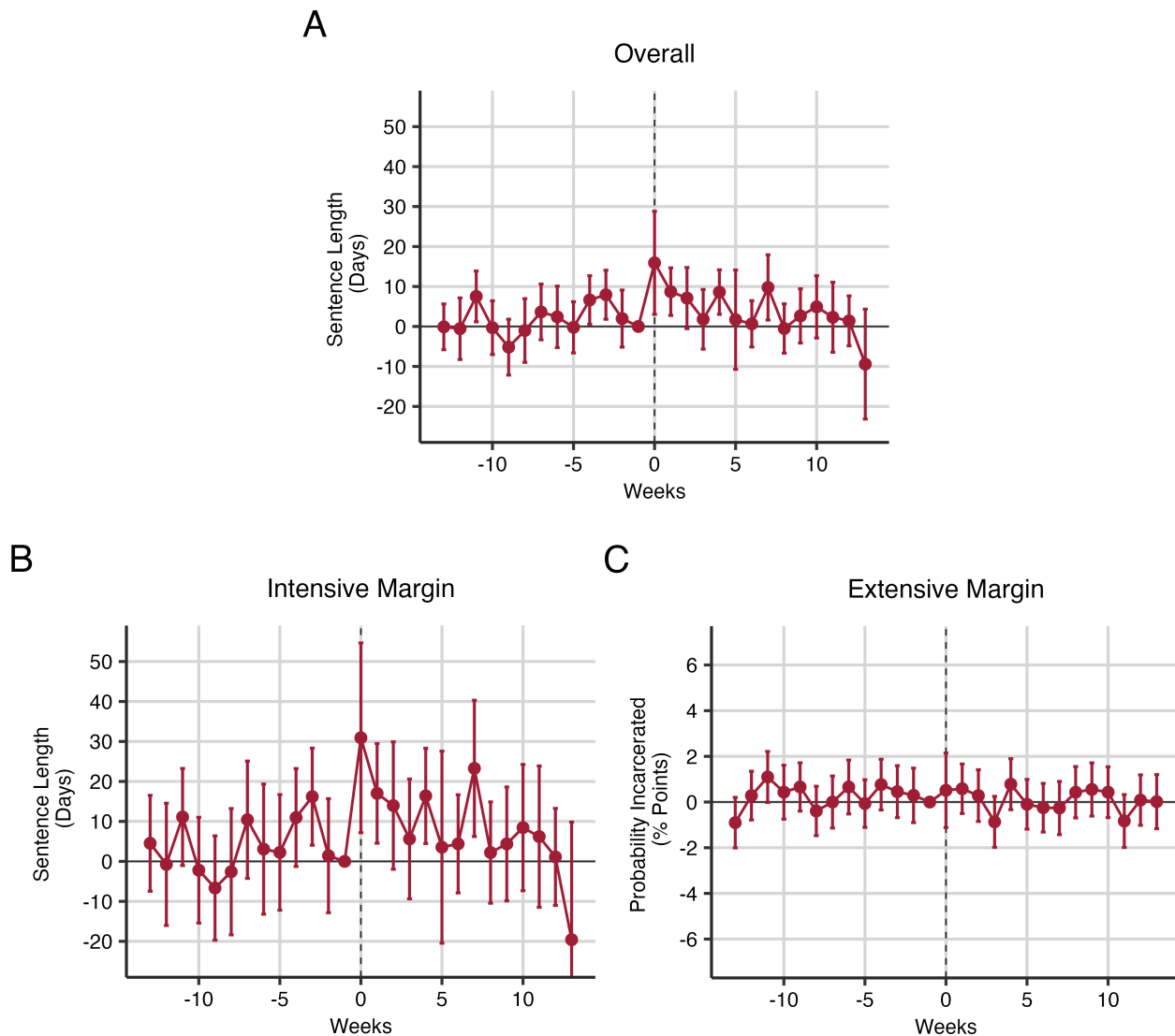
B

Minority Defendants – White Defendants



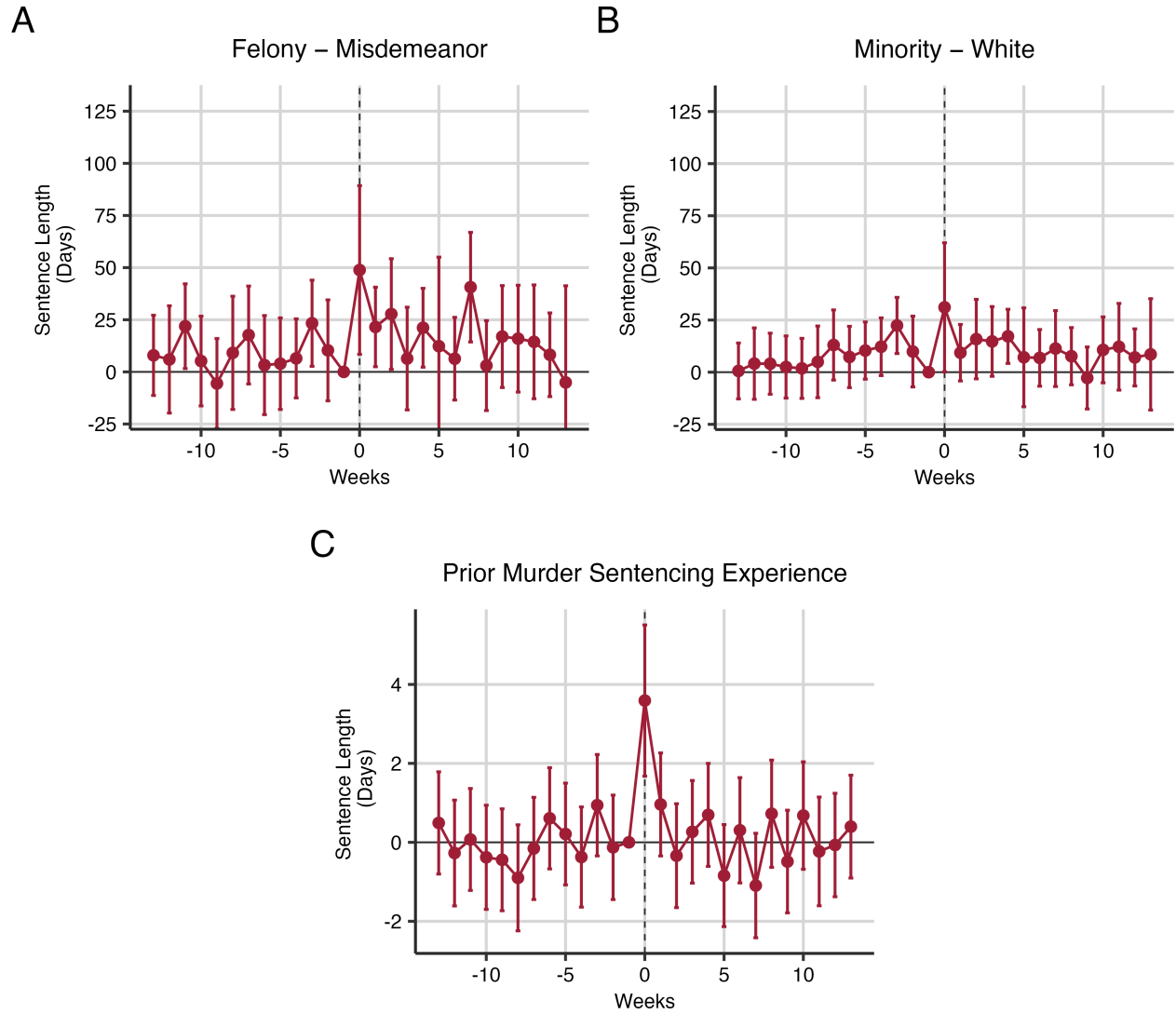
Note: Each panel in this figure plots estimates from a triple-differences design. Time 0 is the day of the murder sentencing. All other points represent weeks in relative time. Bars are 95% confidence intervals. Panel A plots the difference in treatment effects between high-class (X or 1) and low-class (2, 3, or 4) felony charges. Panel B plots the difference in treatment effects between minority and White defendants. Regressions include controls for the initial felony charge class, the final felony charge class, the defendant's race, gender, and age, and fixed effects for the sentencing judge and the murder sentencing event.

Figure 5: Average Treatment Effects and Margin Decomposition (Pennsylvania)



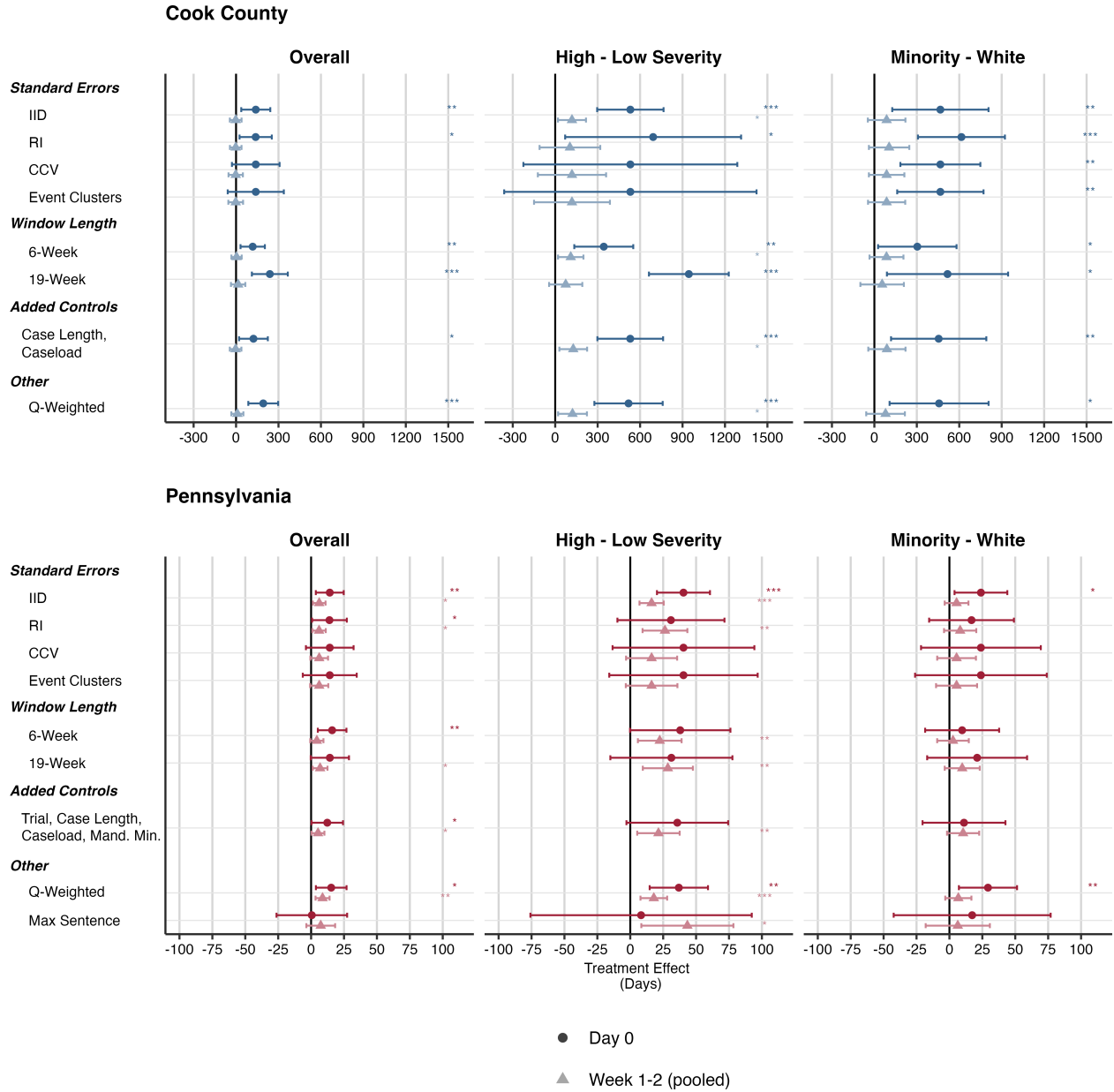
Note: Each panel in this figure plots coefficients from a difference-in-differences design comparing judges who sentence a murder to control judges from the same judicial district. Time 0 is the day of the murder sentencing. All other points represent weeks in relative time. Bars are 95% confidence intervals. Panel A plots sentence length. Panel B plots sentence length conditional on incarceration, and panel C plots incarceration probability. Regressions include controls for the defendant's race, gender, and age, and fixed effects for sentencing guidelines cell ($OGS \times PRS \times Era$), sentencing judge and sentencing event.

Figure 6: Heterogeneous Treatment Effects (Pennsylvania)



Note: Panels A and B plot coefficients from triple-differences designs, while Panel C plots a difference-in-differences design where treatment is interacted with the number of murders a judge has previously sentenced. Time 0 is the day of the murder sentencing. All other points represent weeks in relative time. Bars are 95% confidence intervals. Regressions include controls for the defendant's race, gender, and age, and fixed effects for sentencing guidelines cell ($OGS \times PRS \times Era$), sentencing judge and sentencing event.

Figure 7: Robustness (Cook County/Pennsylvania)



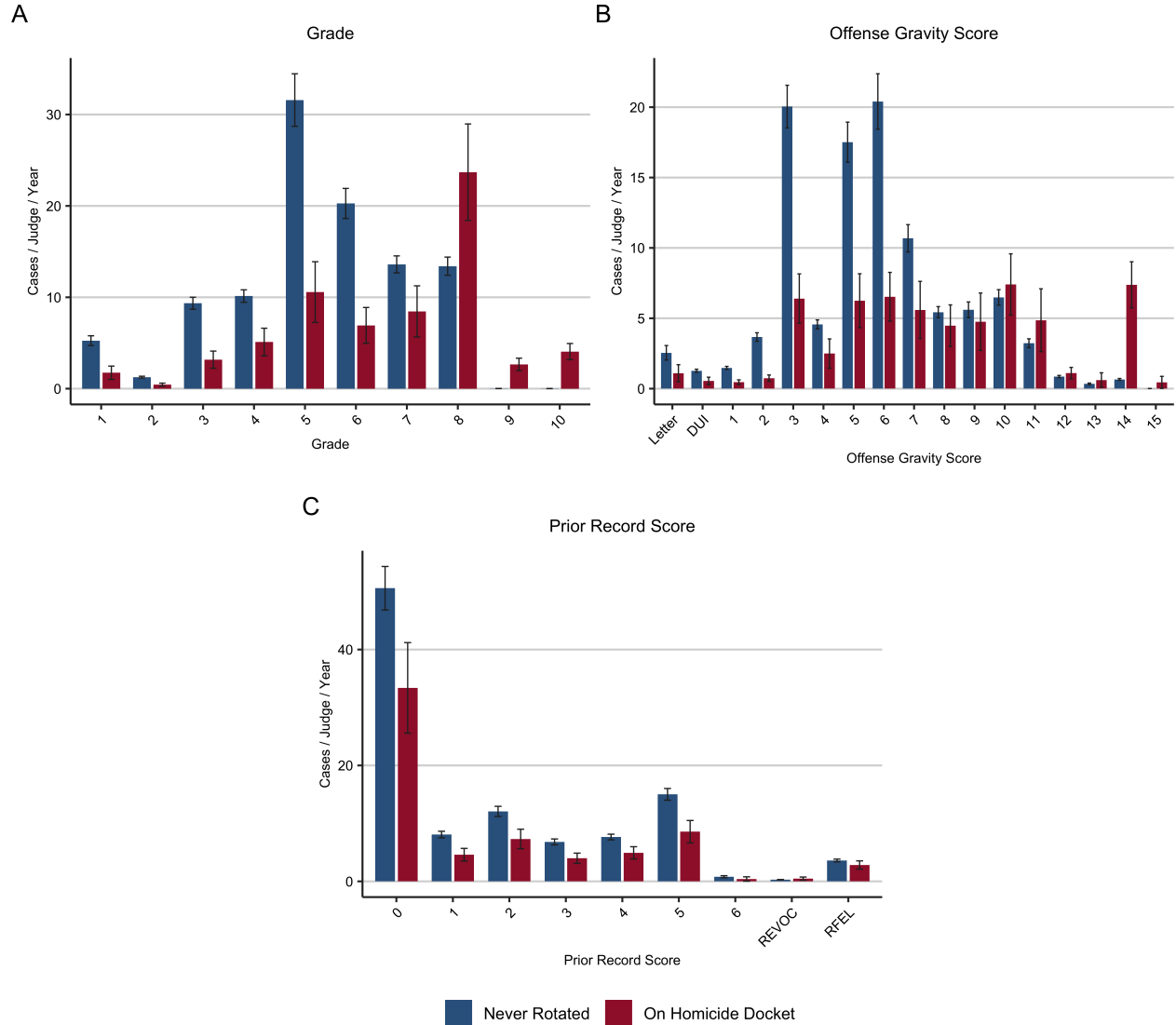
Note: Circles plot the murder sentence day effect (day-0 coefficient minus pre-period mean). Triangles plot the pooled effect during the two weeks afterward (the average of the week-1 and week-2 coefficients minus the pre-period mean). Bars are 95% confidence intervals. Stars denote significance at the 5% (*), 1% (**), and 0.1% (***) levels. The *Standard Errors* rows estimate IID, RI, CCV, and event-clustered standard errors. The *Window Length* rows re-estimate using alternate event windows and pre-period restrictions. The *Controls* row adds case length and judge caseload in both jurisdictions, plus trial conviction and mandatory-minimum flags in Pennsylvania. *Q-Weighted* re-weights control cases to match the distribution of treated cases across events (Wing et al., 2024). *Max Sentence* replaces the min sentence outcome with the max.

Figure 8: Philadelphia Homicide Docket Variation



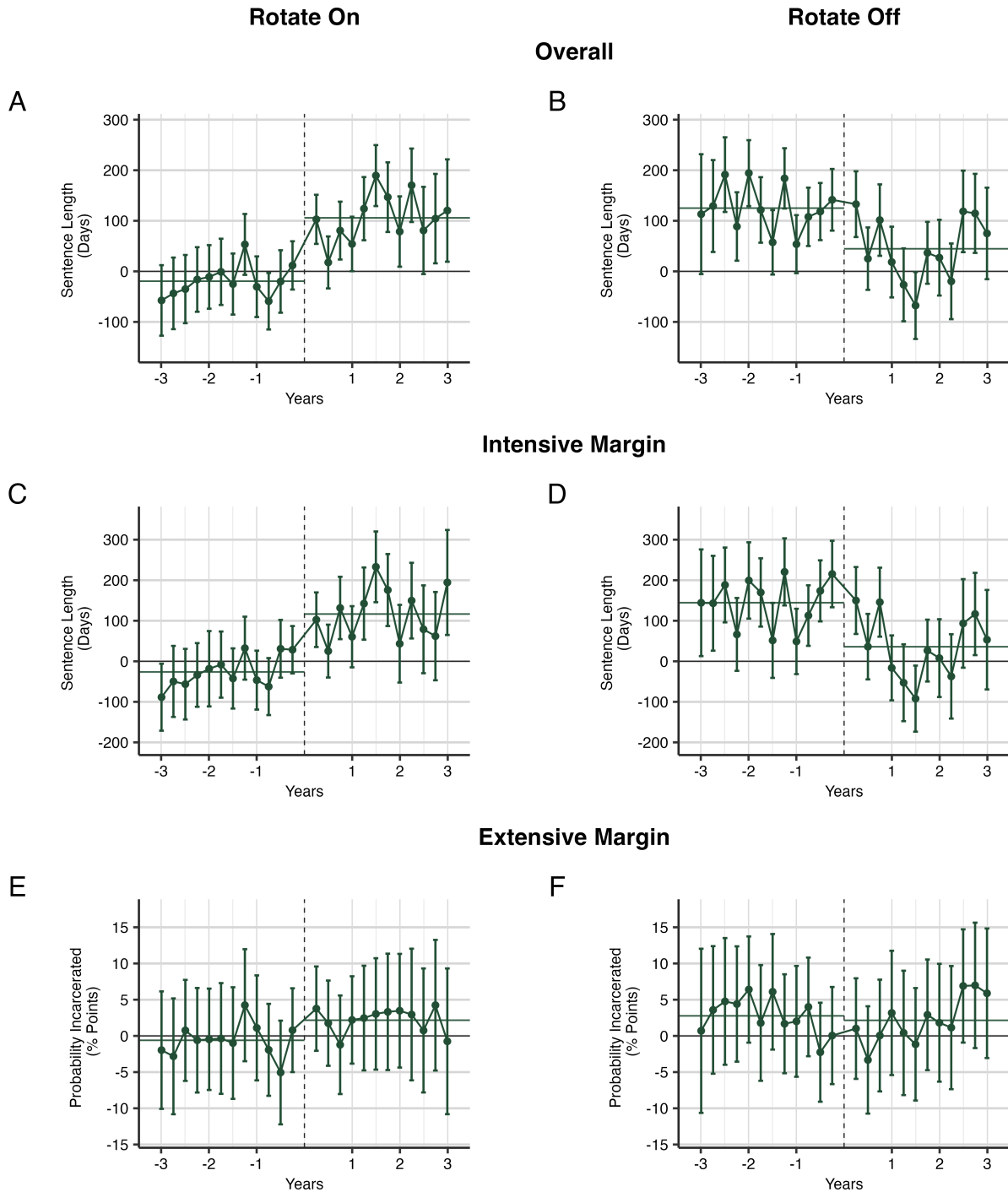
Note: This figure documents detected homicide-docket rotations in the Philadelphia Court of Common Pleas. Each horizontal line corresponds to one judge, ordered vertically by the date of their first observed case. Gray segments show a judge's full active career period. Green segments highlight periods identified as homicide-rotation windows: intervals bounded by clusters of at least four distinct murder-sentencing dates within one year of each other. Dark green points mark individual murder-sentencing dates. Judges without a detected rotation are omitted.

Figure 9: Case Composition of Rotator and Never-Rotated Judges



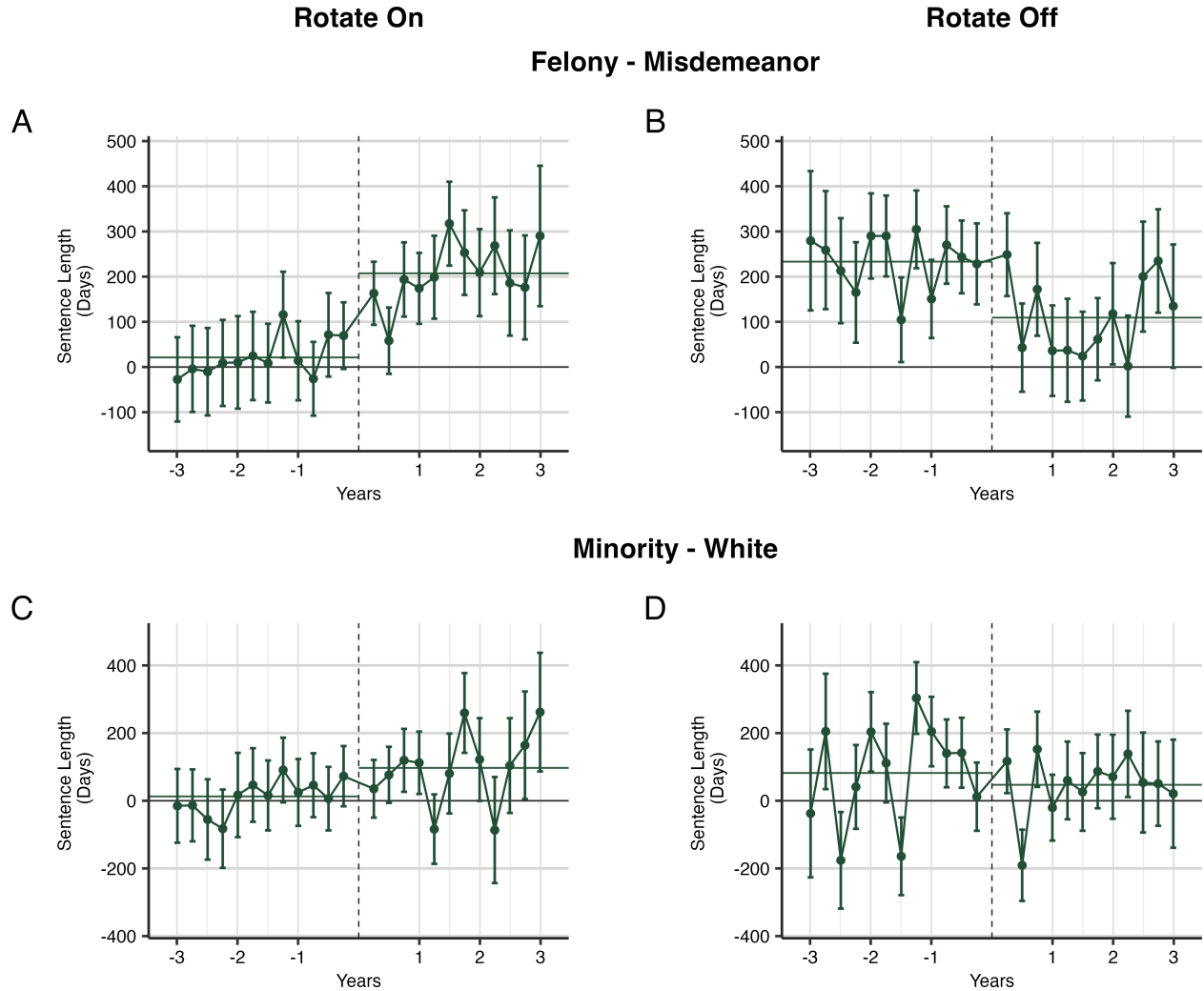
Note: This figure compares the mix of cases sentenced by rotator judges while on the homicide docket to those of never-rotated judges over the same periods. Treated cases are those sentenced by a rotating judge during a rotation window; control cases are those sentenced by never-rotated judges during the same windows. Each bar reports cases per judge per year. Error bars are 95% confidence intervals. Panel A groups cases by charge grade. Panel B groups cases by Offense Gravity Score (“Letter” collects pre-1997 guideline-edition scores and “DUI” collects DUI-specific scores). Panel C groups cases by Prior Record Score.

Figure 10: Effect of Homicide Docket Assignment (Philadelphia)



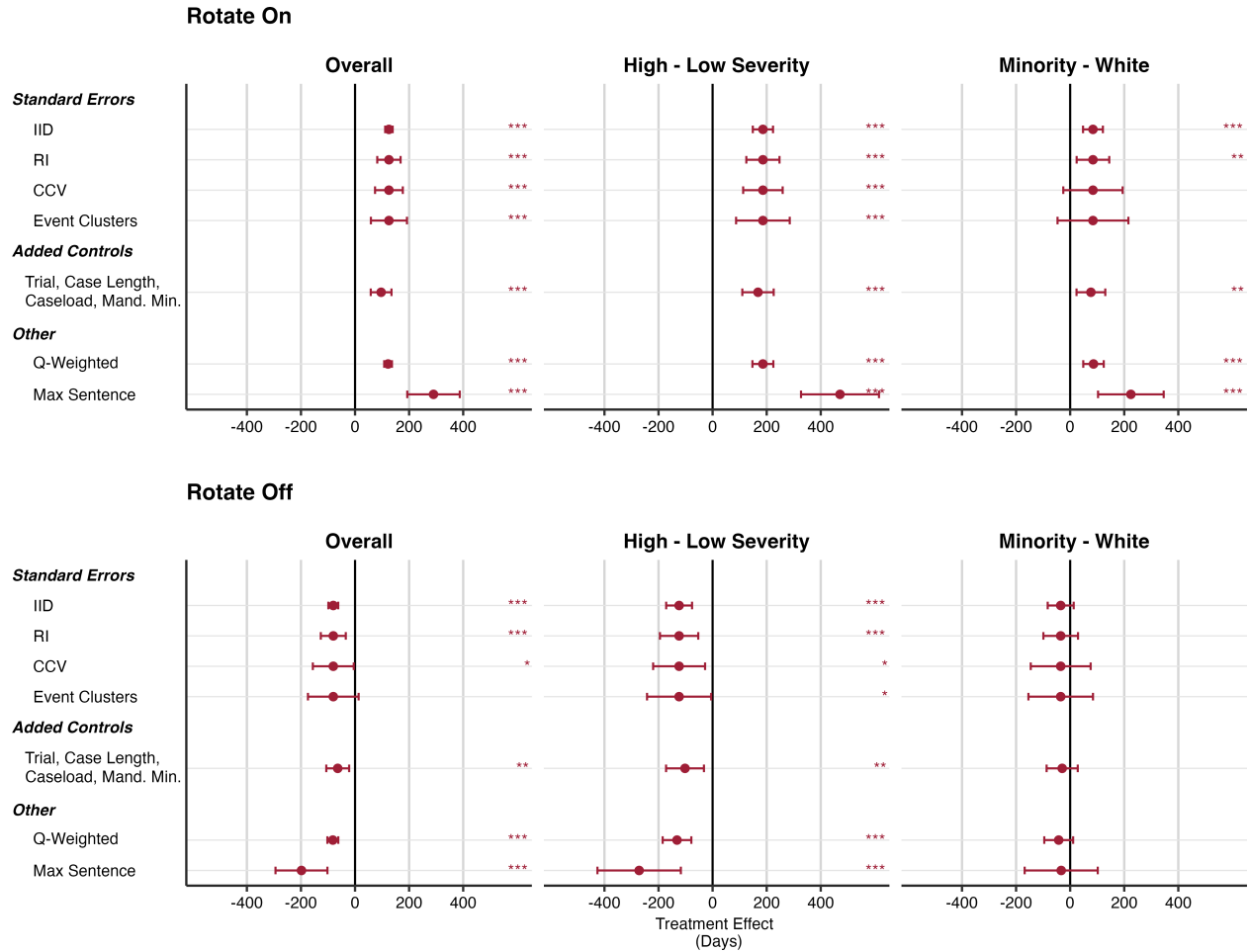
Note: Each panel in this figure plots coefficients from a difference-in-differences design comparing judges who rotate onto (left-hand side) or off of (right-hand side) assignment to the Philadelphia homicide docket to control judges who are never assigned to the Philadelphia homicide docket. Each point represents a quarter in relative time. Bars are 95% confidence intervals. Horizontal lines represent the pre- and post-rotation means. The outcome variable is residualized sentence length with controls for race, age, gender, sentencing guidelines cell ($OGS \times PRS \times Era$), and judge fixed effects.

Figure 11: Homicide Docket Effects: Margins and Heterogeneity (Philadelphia)



Note: Each panel plots coefficients from triple-differences designs. Panels A and B compare effects on felony and misdemeanor sentencing. Panels C and D compare effects on minority and White defendants. Each point represents a quarter in relative time. Bars are 95% confidence intervals. Horizontal lines represent the pre- and post-rotation means. The outcome variable is residualized sentence length with controls for race, age, gender, sentencing guidelines cell ($OGS \times PRS \times Era$), and judge fixed effects.

Figure 12: Robustness (Philadelphia)



Note: The outcome is residualized sentence length with controls for race, age, gender, guidelines cell, and judge fixed effects. Each point is the difference between coefficients for three years of pre-rotation and rotation (rotate-on) or rotation and post-rotation (rotate-off). Bars are 95% confidence intervals. Stars denote significance at the 5% (*), 1% (**), and 0.1% (***) levels. *Standard Errors* rows estimate IID, RI, CCV, and event-clustered standard errors. The *Controls* row adds trial conviction flags, case length, judge caseload, and mandatory-minimum flags. *Q-Weighted* re-weights control cases to match the distribution of treated cases across events (Wing et al., 2024). *Max Sentence* replaces the min sentence outcome with the max.